AIR QUALITY AND POLLUTION ANALYSIS

Course: Data Visualization

Course Code: CAP483

Date: 18/09/2025

Submitted by:

NAME	REG.NO.	ROLL
Jasmeet	12301352	12
Shashi	12301625	13
Shivam	12304068	19

Submitted to:

Anjana Sharma

SCHOOL OF COMPUTER APPLICATIONS
Lovely Professional University, Punjab

1. Acknowledgment
2. Abstract
3. Introduction
4. Objectives of the Study
5. Literature Review (Brief Background)
6. Methodology
    1. Data Collection
    2. Tools Used (Tableau, Excel)
    3. Visualization Techniques
7. Data Analysis and Visualization
    1. Spatial Distribution of Pollution
    2. Top Polluted Cities
    3. Regional Pollution Levels
    4. Dominant Pollutants
8. Dashboard Creation
9. Results and Discussion
10. Findings and Insights
11. Conclusion
12. Recommendations
13. References

Acknowledgment

I take this opportunity to express my sincere gratitude to all those who have contributed to the successful completion of this project titled "Air Quality and Pollution Analysis."

First and foremost, I would like to convey my deepest appreciation to my project guide and faculty mentors for their continuous guidance, constructive feedback, and encouragement throughout every stage of this study. Their expertise and insightful suggestions have been invaluable in shaping the analytical direction and overall quality of this report.

I am also profoundly thankful to the Central Pollution Control Board (CPCB) for making available authentic and comprehensive air quality data, which formed the core foundation of this research. The data provided by CPCB enabled an evidence-based analysis and added credibility to the study's findings.

My heartfelt thanks also go to my department and institution for providing the necessary academic infrastructure, resources, and an intellectually stimulating environment to carry out this research work effectively.

Lastly, I extend my appreciation to my peers, friends, and family members for their constant motivation, patience, and moral support. Their encouragement has played a vital role in helping me complete this project successfully and on time.

Abstract

Air pollution represents one of the most critical environmental challenges of the modern era, with significant implications for human health, climate stability, and ecological balance. This study focuses on analyzing and visualizing air quality data across India to identify pollution hotspots, dominant pollutants, and regional trends. The research utilizes data obtained from the Central Pollution Control Board (CPCB) and employs Tableau and Microsoft Excel for data processing, visualization, and interpretation.

Through various visualization techniques—including geographical maps, bar charts, and treemaps—the study effectively compares pollution levels across cities and states. The analysis reveals that northern cities such as Delhi and Kanpur exhibit the highest concentrations of PM2.5, while southern states like Kerala and Tamil Nadu maintain comparatively cleaner air quality.

The findings emphasize the effectiveness of visualization tools in transforming complex environmental datasets into comprehensible insights. This approach enhances public awareness, facilitates data-driven decision-making, and supports policymakers in formulating more targeted and sustainable air quality management strategies.

Introduction

Air pollution refers to the presence of harmful or excessive substances in the atmosphere, such as gases, particulate matter, and biological molecules, which can cause serious damage to human health and the environment. In India, air pollution has become one of the most concerning issues in recent years due to rapid urbanization, industrial growth, increasing vehicle emissions, and agricultural activities like crop residue burning.

Pollutants such as PM2.5, PM10, CO_2, NO_2, and SO_2 play a major role in the deterioration of air quality. These pollutants not only reduce visibility and affect the environment but also lead to various health problems, including respiratory diseases, asthma, and cardiovascular issues. The problem is more severe in urban and industrial regions, where population density and economic activity are higher.

This project mainly focuses on understanding and analyzing the air pollution situation across different parts of India. Using data visualization techniques, the aim is to identify pollution trends, regional variations, and key pollutants responsible for poor air quality in specific areas. By using tools like Tableau and Excel, complex air quality data can be represented in a clear and meaningful way, which helps in understanding patterns more easily.

Through this analysis, we can identify which regions are most affected, which pollutants are dominant, and how visualization can be used to support better decision-making and awareness about environmental issues. The project also highlights the importance of taking necessary actions to reduce pollution and protect public health for a sustainable future.

Objectives of the Study

The main purpose of this project is to study the air quality across India using data visualization techniques. By analyzing data from different cities and states, the project aims to understand pollution patterns and the factors influencing them. The specific objectives of this study are as follows:

1. To identify major air pollutants such as PM2.5, PM10, CO_2, NO_2, and SO_2, which contribute significantly to air quality degradation.
2. To compare pollution levels across different cities and states in India, in order to find out which regions are most affected.
3. To analyze spatial patterns of air pollution using visualization tools, helping to identify pollution hotspots and regional trends.

4. To understand the causes and possible health impacts of high pollution levels on people and the environment.
5. To create an interactive and user-friendly dashboard using Tableau, which allows easy interpretation of data and comparison between different pollutants and locations.

These objectives together help in gaining a clearer understanding of India's air quality status and how visualization can make complex data more meaningful and accessible.

Literature Review (Brief Background)

Several studies and reports by national and international organizations have highlighted that air pollution is one of the most serious environmental issues in India. According to the World Health Organization (WHO) and the Central Pollution Control Board (CPCB), many Indian cities are among the most polluted in the world. The presence of fine particulate matter such as PM2.5 and PM10 is a major concern, as these particles can penetrate deep into the lungs and bloodstream, causing respiratory, cardiovascular, and even neurological diseases.

Research findings have shown that cities like Delhi, Kanpur, Lucknow, Varanasi, and Patna regularly record high levels of air pollution. The main reasons behind this are rapid urbanization, industrial activities, vehicular emissions, and burning of crop residues in nearby agricultural regions. The high population density and lack of proper urban planning further worsen air quality, especially during the winter season when weather conditions trap pollutants near the ground.

On the other hand, several studies indicate that southern states such as Kerala, Tamil Nadu, and Karnataka have comparatively better air quality. This is often attributed to stronger environmental regulations, higher green cover, and favorable coastal wind patterns that help in dispersing pollutants more effectively.

Past research has mainly focused on measuring pollution levels and identifying sources. However, there has been less emphasis on presenting this information in an easily understandable way. This project aims to bridge that gap by using data visualization tools like Tableau to make pollution data more interactive and insightful. By visualizing the data through maps, charts, and dashboards, this study helps in simplifying complex datasets and provides a clearer understanding of how air pollution varies across different regions of India.

Methodology

This project is based on secondary data analysis, which means the data was collected from reliable and verified government sources rather than through direct measurement or fieldwork. The study mainly uses data published by the Central Pollution Control Board (CPCB), which regularly monitors air quality across various Indian cities and states.

The overall methodology focuses on three main stages — data preparation, data visualization, and data interpretation using modern analytical tools like Microsoft Excel and Tableau.

In the data preparation stage, the dataset was first examined, cleaned, and organized in Excel to remove any inconsistencies or missing values. Columns such as city, state, pollutant type, pollutant average, latitude, and longitude were formatted properly for easy import into Tableau.

In the visualization stage, Tableau was used to create different types of visual representations such as maps, bar charts, treemaps, and dashboards. These visualizations helped to clearly identify pollution hotspots, compare pollution levels across regions, and determine the dominance of specific pollutants.

Finally, in the interpretation stage, the generated visuals were analyzed to draw meaningful insights about the air quality trends across India. This included understanding which areas are most affected, which pollutants are most dominant, and how pollution varies between urban and rural or industrial regions.

The use of visualization tools not only made the analysis more interactive but also helped in transforming raw numerical data into meaningful information that can assist in decision-making and public awareness.

Data Collection

The data used in this project was collected from the Central Pollution Control Board (CPCB), which is India's primary government agency responsible for monitoring and controlling pollution levels across the country. The CPCB continuously records air quality data from various monitoring stations located in major cities and states.

- Source: Central Pollution Control Board (CPCB), India
- Dataset Type: Excel file (Air Pollution Dataset)
- Date Collected: 28 October 2025
- Parameters Included: City, State, Pollutant Type, Pollutant Average, Station ID, Latitude, Longitude

The dataset contains information from numerous air quality monitoring stations spread across different regions of India. Each record includes readings for various pollutants such as PM2.5, PM10, CO_2, NO_2, and SO_2. These pollutants are measured in micrograms per cubic meter ($\mu g/m^3$), which allows for consistent comparison across locations.

This dataset was selected because it is authentic, regularly updated, and publicly available through government portals. It provides sufficient coverage of both urban and industrial regions, which makes it possible to perform a detailed comparative analysis between different cities and states.

By using this dataset, the study was able to observe how pollution levels vary geographically, identify pollution hotspots, and understand which pollutants are most dominant in particular regions. The data serves as a strong foundation for visualization and further interpretation using Tableau.

Tools Used (Tableau, Excel)

Two main software tools were used in this project — Microsoft Excel and Tableau. Both tools played an important role in organizing, analyzing, and visualizing the air quality data collected from the CPCB.

Microsoft Excel

Microsoft Excel was primarily used for the initial data cleaning and preparation process. Since the dataset contained several parameters such as pollutant type, average pollutant value, city, and state, Excel helped in:

- Removing blank cells and duplicate entries.
- Sorting and filtering data based on pollutants and regions.
- Formatting numeric and text values for compatibility with Tableau.
- Performing simple statistical calculations such as averages and totals.

By cleaning and organizing the data in Excel first, it became easier to import it into Tableau without errors or inconsistencies.

Tableau

Tableau was used as the main data visualization tool for this project. It helped transform large volumes of raw data into clear, meaningful, and visually appealing charts and maps. Using Tableau, different types of visualizations were created, such as:

- Maps: To show the geographical distribution of pollution levels across India.

- Bar Charts: To rank cities based on pollution levels and identify the most affected areas.
- Treemaps: To compare pollution levels across different states.
- Dashboards: To combine multiple visualizations into one interactive and user-friendly interface.

Tableau's interactive features such as filters, tooltips, and color coding made the visualizations easy to explore and interpret. It allowed for quick comparisons between pollutants, cities, and states, helping to highlight key insights about India's air quality more effectively than traditional data tables or reports.

Visualization Techniques

To make the air pollution data easier to understand and analyze, four main types of visualizations were created using **Tableau**. Each visualization was designed to highlight important trends, patterns, and comparisons between different regions and pollutants. These visualizations helped convert large datasets into clear and interactive visuals that support better understanding and decision-making.

1. **Spatial Map Visualization**

The spatial map was created to show the geographical distribution of pollutants across India. Using the latitude and longitude values from the dataset, each city was represented as a bubble on the map. The size and color intensity of the bubbles indicated the average concentration of pollutants — larger and darker bubbles showed higher pollution levels.
This type of visualization helped in identifying pollution hotspots, especially in northern and industrial regions like Delhi, Kanpur, and Lucknow. It also showed that southern parts of India generally had cleaner air.

2. **Bar Chart Visualization**

The bar chart was used to display the top 10 most polluted cities based on the average concentration of pollutants. Each bar represented a city, with the length showing the pollution level. The bars were color-coded from green (low pollution) to red (high pollution) to make comparison easier at a glance.
This visualization helped in quickly identifying the cities with the worst air quality and allowed for an easy comparison between urban areas.

3. **Treemap Visualization**

The treemap visualization compared pollution levels across different Indian states. In this chart, each block represented a state, and the size and color of the block indicated its average pollution level. States like Maharashtra, Uttar Pradesh, and Gujarat appeared as larger and darker blocks, showing higher levels of pollution, while states

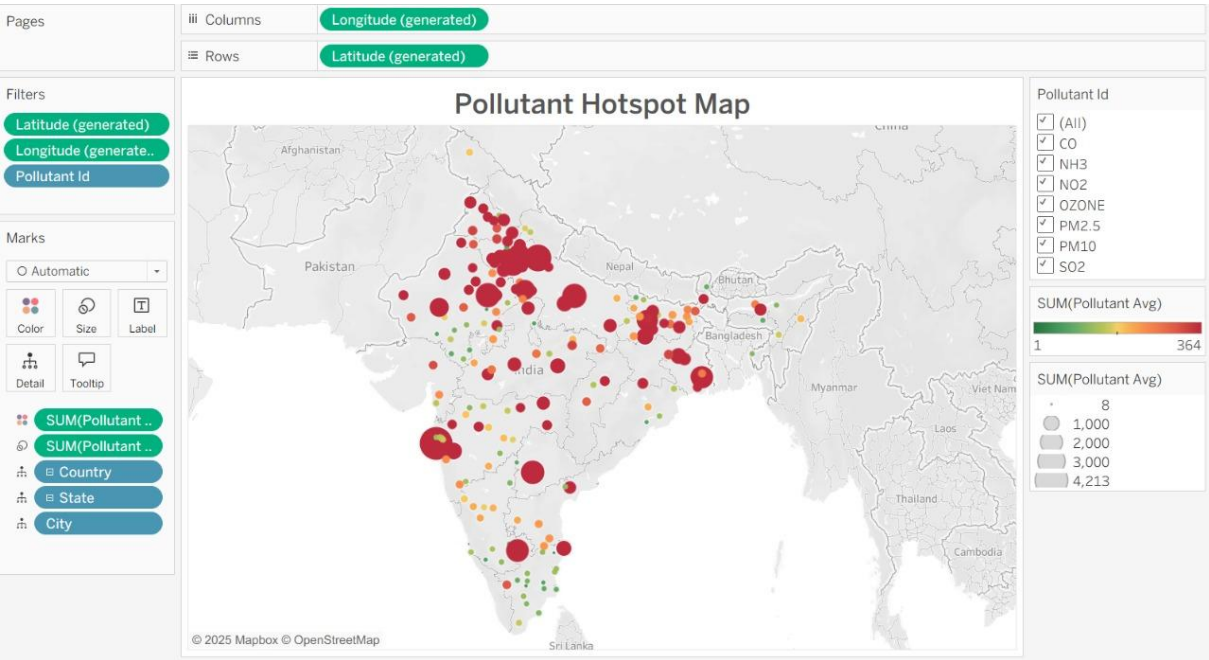
like Kerala and Tamil Nadu appeared smaller and lighter, indicating better air quality. This visualization made it simple to compare pollution on a regional level and identify which states require stricter environmental measures.

4. **Pollutant Profile Chart**

The pollutant profile chart showed which pollutants are most dominant in specific cities or regions. It used vertical bars to represent pollutants such as PM2.5, PM10, CO_2, NO_2, and SO_2, with each bar's height showing its average level. Different colors were used for each pollutant category to make the chart more visually appealing and easy to interpret. This chart helped in understanding the composition of air pollution — for example, identifying that PM2.5 was the most dominant pollutant in most northern cities, while CO_2 was higher in industrial states.

Data Analysis and Visualization

Spatial Distribution of Pollution



To understand how air pollution varies across different parts of India, a geographical map was created using Tableau. The map was designed using the longitude and latitude coordinates from the dataset to plot the locations of air quality monitoring stations across the country. Each city or station appeared as a bubble on the map, where both the size and color of the bubble represented the average pollution level (Pollutant Avg).

In this visualization, darker and larger bubbles indicated higher concentrations of pollutants, while smaller and lighter bubbles represented areas with lower pollution levels. The color scale ranged from green (cleaner air) to red (high pollution), making it easy to identify the regions that were most affected.

The map revealed a clear pattern — northern and central India had significantly higher pollution levels compared to other regions. Cities like Delhi, Kanpur, Lucknow, Varanasi, and

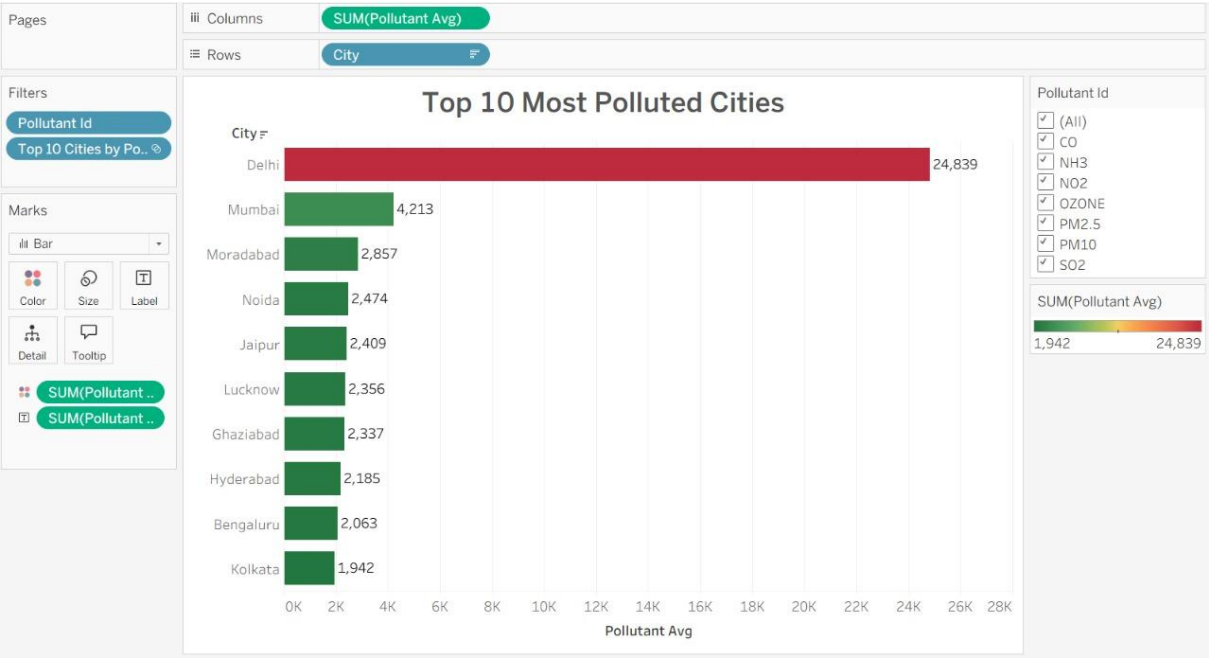
Patna showed severe pollution, mainly due to heavy traffic, industrial activities, and crop residue burning during certain seasons. These regions consistently recorded high concentrations of PM2.5 and PM10, which are the most harmful particulate pollutants.

In contrast, southern India, particularly states such as Kerala, Tamil Nadu, and Karnataka, displayed lighter colors on the map, indicating relatively cleaner air. This difference is likely due to better air circulation, coastal winds, and stricter pollution control measures in those states.

The spatial distribution map thus helped visualize pollution patterns across India at a glance. It highlighted where pollution control efforts need to be strengthened and which regions are performing better in maintaining cleaner air. Such a map also makes it

easier for policymakers and environmental agencies to focus on specific high-risk zones for targeted intervention.

Top Polluted Cities



To identify the most affected urban areas, a bar chart visualization was created in Tableau showing the top 10 most polluted cities in India. The cities were ranked according to their average pollutant concentration values taken from the dataset. Each bar represented a city, with its length corresponding to the pollution level.

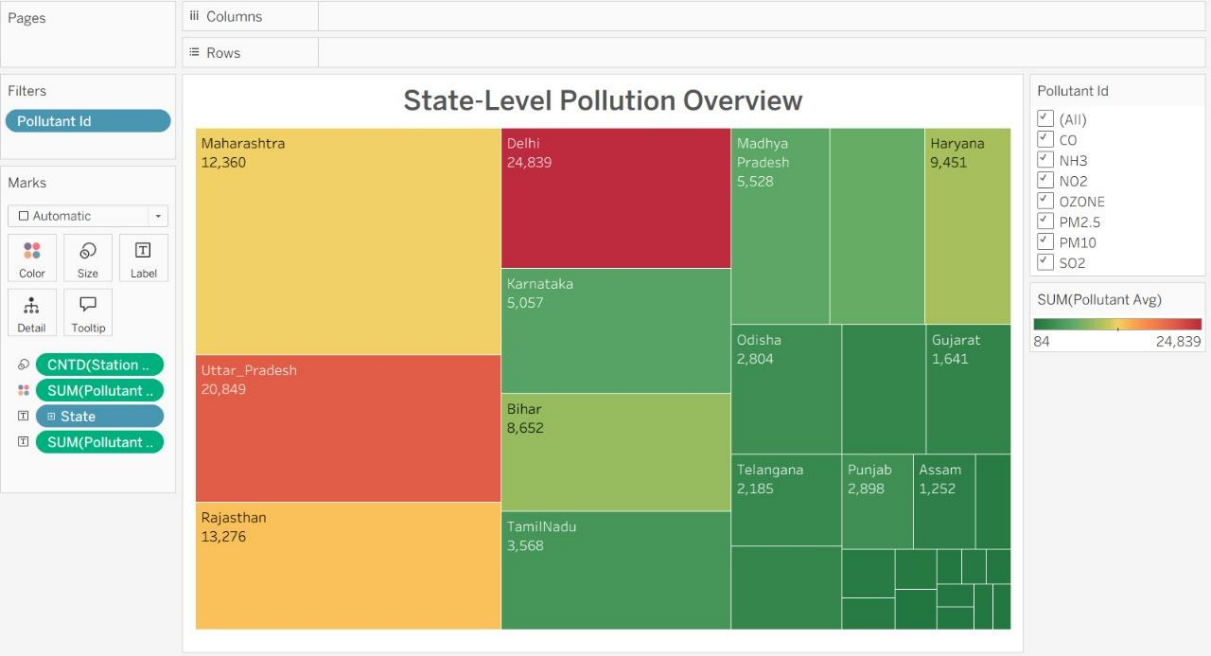
The color scheme ranged from green to red, where green indicated low pollution and red represented very high pollution. This color gradient made it easier to compare cities visually and quickly understand their air quality levels.

The analysis revealed that cities such as Delhi, Kanpur, and Lucknow consistently recorded the highest pollution concentrations, mainly due to high population density, vehicle emissions, and industrial activity. Other northern cities like Varanasi, Patna, and Ghaziabad also appeared near the top of the list, indicating poor air quality conditions across the northern belt of India.

Among the pollutants, PM2.5 was found to be the most dominant, followed by PM10. These fine particulate matters are especially harmful because they can enter deep into the lungs and cause serious respiratory and cardiovascular diseases. The bar chart not only highlighted the cities that need immediate attention but also provided a clear visual comparison between regions with different air quality conditions.

This visualization demonstrated that urbanization and industrial growth have a direct impact on pollution levels. It also emphasized the need for strict emission control measures and better city-level environmental policies to reduce pollution in highly affected areas.

Regional Pollution Levels



To analyze the data on a broader scale, a treemap visualization was created to compare pollution levels across various Indian states. In this visualization, each state was represented as a rectangular block. The size and color intensity of each block corresponded to its average pollution level — larger and darker blocks indicated higher pollution, while smaller and lighter blocks showed cleaner air.

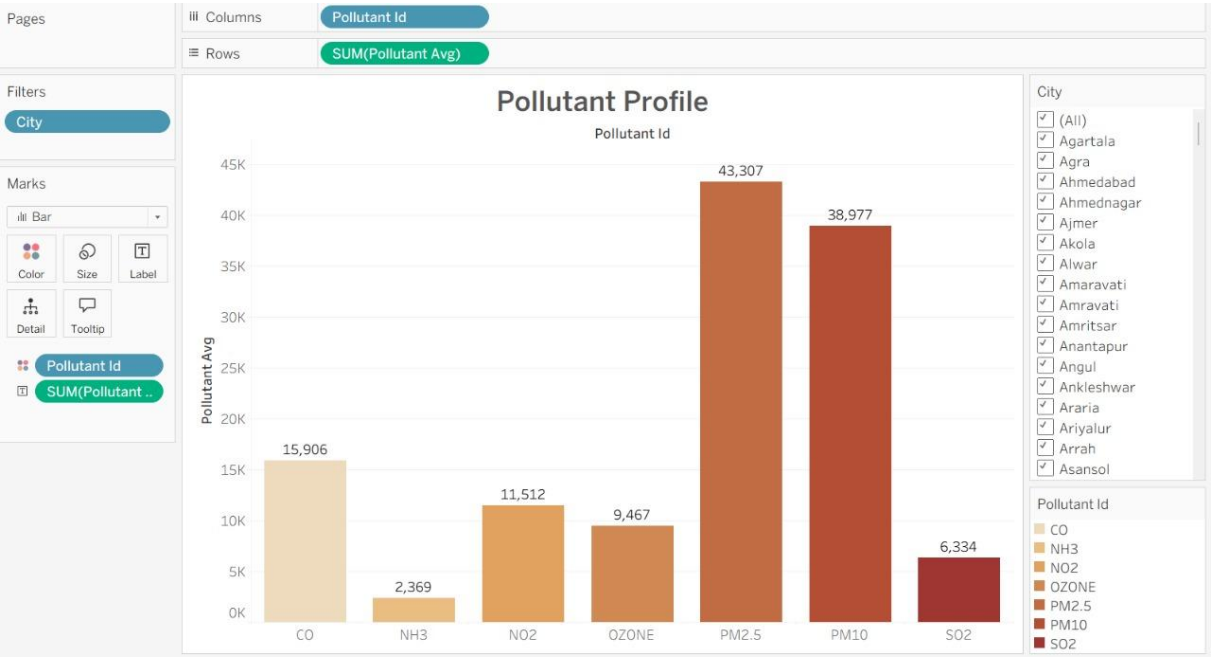
From the treemap, it was observed that Maharashtra, Uttar Pradesh, and Gujarat had higher pollution levels. This can be linked to the large number of industrial zones, dense populations, and vehicle usage in these states. These regions also have several urban

centers with significant traffic congestion, contributing further to air quality degradation.

In contrast, southern states like Kerala and Tamil Nadu appeared with smaller and lighter-colored blocks, representing lower pollutant averages. These states benefit from factors such as coastal wind movement, higher rainfall, and better enforcement of environmental policies, which help in maintaining cleaner air quality.

This regional comparison made it easier to see the difference in pollution intensity between industrial and non-industrial regions. The treemap provided a clear picture of how pollution varies across states and helped identify where policy interventions and air quality improvement programs are most needed.

Dominant Pollutants



To understand which pollutants contribute the most to poor air quality, a pollutant profile chart was created using Tableau. This chart compared the average levels of different pollutants such as PM2.5, PM10, CO_2, NO_2, and SO_2 across various cities and states in India. Each pollutant was represented by a different color bar, and the height of the bar indicated its average concentration.

From the analysis, it was found that PM2.5 (Particulate Matter 2.5) was the most dominant pollutant in most regions of India, especially in large cities like Delhi, Kanpur, Lucknow, and Varanasi. PM2.5 consists of very fine particles that remain suspended in the air for a long time and can easily enter the lungs, causing breathing difficulties, asthma, and other serious health problems.

The second most common pollutant was PM10, which includes slightly larger dust particles mainly generated from road dust, construction work, and industrial processes. Together, PM2.5 and PM10 contribute the most to the degradation of air quality across the country.

In addition to particulate matter, CO_2 (carbon dioxide) levels were observed to be higher in industrial states such as Maharashtra and Gujarat, mainly due to manufacturing activities and fossil fuel combustion. NO_2 (nitrogen dioxide) concentrations were also significant in high-traffic areas, as it is a common byproduct of vehicle emissions. SO_2 (sulfur dioxide) was found in moderate amounts, primarily released from thermal power plants and industries using coal or oil.

This visualization clearly showed that different pollutants dominate in different regions, depending on local sources of pollution. For example, industrial areas tend to have higher CO_2 and SO_2 levels, while densely populated cities suffer mainly from PM2.5 and PM10 polluon.

Understanding which pollutants are dominant in each area helps policymakers and researchers focus on specific pollution sources and implement more targeted control measures to improve air quality.

Dashboard Creation

After creating all the individual visualizations such as maps, bar charts, treemaps, and pollutant profile charts, an interactive dashboard was designed in Tableau to bring all the elements together in one place. The dashboard helps users view and analyze multiple aspects of air pollution data at the same time, making the study more engaging and insightful.

The dashboard included the following key components:

- Hotspot Map: Displaying the spatial distribution of pollutants across India, showing where air pollution is most concentrated.
- Top Cities Chart: Listing the top 10 most polluted cities, allowing quick comparison between urban areas.
- State-Level Treemap: Showing regional differences in pollution levels, helping identify the most and least affected states.
- Pollutant Profile Chart: Highlighting which pollutants are dominant in specific cities or regions.

Each visualization was carefully arranged to maintain a clean and organized layout. The maps were placed at the top of the dashboard to give an immediate overview of the national air quality scenario, while the bar and treemap charts were positioned below to provide detailed comparisons.

Interactive filters were also added to make the dashboard more dynamic. Users could filter data by pollutant type, state, or city to focus on specific areas of interest. For instance, selecting PM2.5 from the filter instantly updated all charts to display only PM2.5-related data.

Additionally, Key Performance Indicators (KPIs) such as the Highest Polluting City (Delhi), Number of Reporting Stations (483), and the National Average Pollution Level (41.22) were displayed at the top of the dashboard. These KPIs provided quick summary information for better understanding and reporting.

The final dashboard served as an effective tool for presenting complex data in a visually simple and interactive format. It allowed for quick insights into pollution patterns, regional comparisons, and pollutant dominance across India. This type of visualization is not only helpful for research and academic work but also for policymakers and environmental organizations that monitor air quality and develop improvement strategies.

Results and Discussion

From the data visualization, we got a clear picture of how air pollution looks across India. Some interesting points came out of the analysis:

- **Delhi and Kanpur had the highest PM2.5 levels.** This shows that these cities have very bad air quality compared to others. PM2.5 is a tiny particle that can go deep into our lungs and cause health problems, so it's worrying to see such high numbers in these cities.
- **Northern regions are more polluted than southern ones.** States like Uttar Pradesh and Delhi have worse pollution, while southern states like Kerala or Tamil Nadu showed lower pollution levels. This could be because the north has more industries, vehicles, and colder weather in winter, which traps pollution near the ground.
- **Main causes of pollution are industries and vehicles.** Factories release smoke and chemicals, and cars give out a lot of exhaust. Together, they make the air dirty. The data shows that places with more factories and traffic tend to have worse air quality.
- **Tableau helped in understanding the data easily.** By using graphs, maps, and color coding, it was simple to spot which areas are most polluted, how pollution changes over time, and if there are unusual spikes in certain cities. It's much easier to understand compared to just looking at numbers in a table.
- **Why this study matters.** Visualization helps turn lots of confusing numbers into something we can quickly understand and use. For example, policymakers can see which cities need urgent attention, or citizens can understand why wearing masks in winter in northern India is necessary.

Findings and Insights

From the analysis, several important points about air pollution in India became clear:

- **PM2.5 is the main problem.** This tiny particle is the biggest threat to air quality because it can get deep into our lungs and even enter the bloodstream. Cities with high PM2.5 levels are at serious risk of health issues like asthma and heart problems.

- **Big cities and industrial areas are worse off.** Places with lots of people, vehicles, and factories, like Delhi and Kanpur, tend to have much higher pollution. This shows that human activity is a major cause of dirty air.
- **Southern states have cleaner air.** States in the south, like Kerala or Karnataka, generally have better air because of stricter environmental rules, fewer industries, and natural factors like more wind and greenery that help disperse pollution.
- **Monitoring and dashboards are very useful.** Using real-time data and visual tools, policymakers can see pollution levels as they change, figure out which areas are in danger, and take action faster. For example, they can restrict traffic or close factories temporarily if pollution spikes.
- **Public awareness is key.** Educating people about the dangers of pollution and how they can help—like using public transport, planting trees, or reducing waste burning—can make a big difference in improving air quality and community health.

Conclusion

- This project showed that data visualization is a powerful way to understand air pollution in India. By looking at the graphs and charts, it became clear that certain cities, especially in the north, have very high pollution levels, and PM2.5 is the most dangerous pollutant affecting people's health.
- Using Tableau made it easier to see patterns and trends in the data. For example, we could quickly identify which cities are the worst affected, how pollution changes over time, and even spot unusual spikes. This kind of visual information is helpful not only for researchers but also for policymakers and the public, so they can take timely action.
- The study also highlights that improving air quality in India will require multiple efforts. Stricter environmental rules, using cleaner energy like solar and wind, and getting people involved—like reducing vehicle use, planting trees, and avoiding burning waste—are all essential steps. With these actions, it's possible to make the air cleaner and healthier for everyone.
- In short, this project proves that combining data analysis with visualization can turn complicated numbers into real insights, helping us understand problems better and find practical solutions.

Recommendations

Based on the study of air pollution in India, several actions can help improve air quality:

- **Expand monitoring networks** – Currently, most pollution monitoring happens in big cities. Setting up sensors in rural and semi-urban areas will give a complete picture of air quality across the country.

- **Encourage electric vehicles** – Reducing the use of petrol and diesel vehicles can cut down harmful emissions. Promoting electric vehicles and public transport can make a big difference in city air quality.
- **Implement strict industrial emission rules** – Factories release a lot of pollutants. Strong regulations and regular checks can ensure industries reduce smoke and chemical emissions.
- **Promote renewable energy** – Using solar, wind, and other clean energy sources instead of coal or petrol can reduce CO_2 emissions and help combat air pollution.
- **Use real-time visualization dashboards** – Tools like Tableau or other dashboards can help government agencies monitor pollution as it happens. This allows quick actions if pollution levels rise suddenly.
- **Raise public awareness** – Educating people about pollution, its health effects, and ways to reduce emissions—like avoiding open waste burning, planting trees, or using public transport—can have a big impact on community health.